Blockchain Exp 7

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Aim: Implement the embedding wallet (Metamask) and transaction using Solidity

Theory:

Q1: What is MetaMask?

MetaMask is a cryptocurrency wallet and browser extension that allows users to interact with blockchain networks such as Ethereum. It enables users to store, send, and receive digital assets securely. MetaMask also acts as a bridge between web browsers and decentralized applications (DApps), allowing users to sign transactions and deploy smart contracts. It provides account management, private key security, and seamless integration with platforms like Remix IDE.

Q2: What is a Test Network (Testnet)?

A test network, or testnet, is a blockchain environment used for testing purposes. It simulates the functionality of the main Ethereum network but uses virtual currency instead of real funds.

Developers use testnets like Sepolia or Goerli to deploy and test smart contracts without financial risk. Testnets help identify bugs, verify functionality, and ensure proper execution before deploying applications on the mainnet.

Q3: Steps to Connect MetaMask with Remix IDE for Performing Transactions

- Install the MetaMask extension from the Chrome Web Store ● Create a new wallet or import an existing wallet using a secret recovery phrase
- Enable test networks in MetaMask settings
- Select a test network such as Sepolia
- Add test ETH using a faucet for performing transactions ●

Open Remix IDE in the browser (https://remix.ethereum.org) ●

Write and compile the smart contract in Remix

- Go to the Deploy & Run Transactions tab
- Select Injected Provider - MetaMask as the environment
- Connect MetaMask account when prompted
- Click Deploy and confirm the transaction in MetaMask ● Interact with deployed contract functions and approve transactions via MetaMask

Implementation:

Step 1: Install and Set Up MetaMask

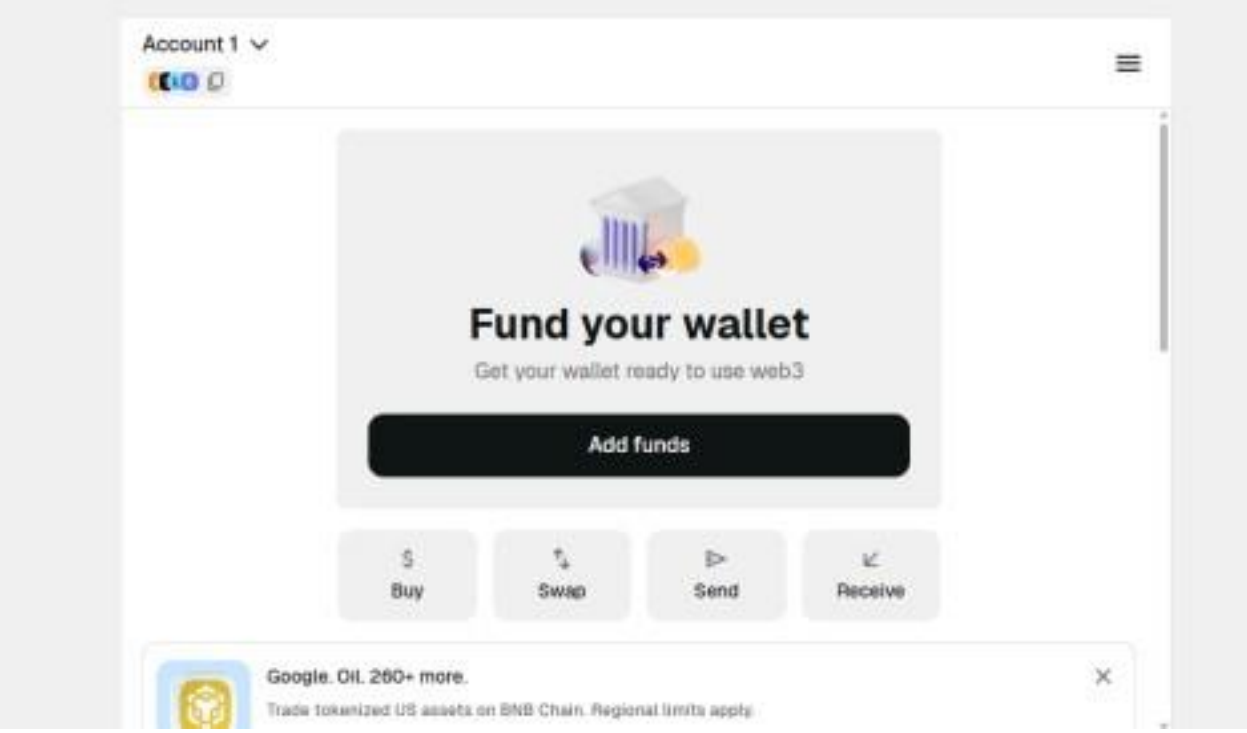
- Install MetaMask extension in Chrome/Brave
- Click “Create a Wallet”
- Set a password

Step 2: Create Landlord and Tenant Accounts

- Open MetaMask
- Click profile icon → **Add Account**

Create:

- Account 1 → Landlord
- Account 2 → Tenant

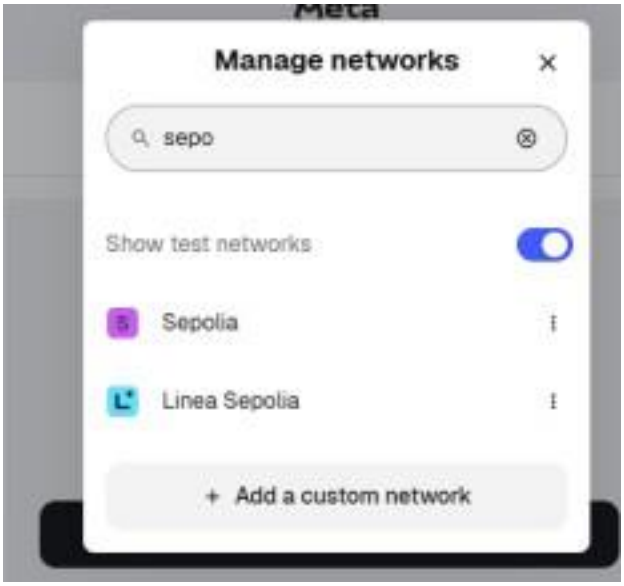


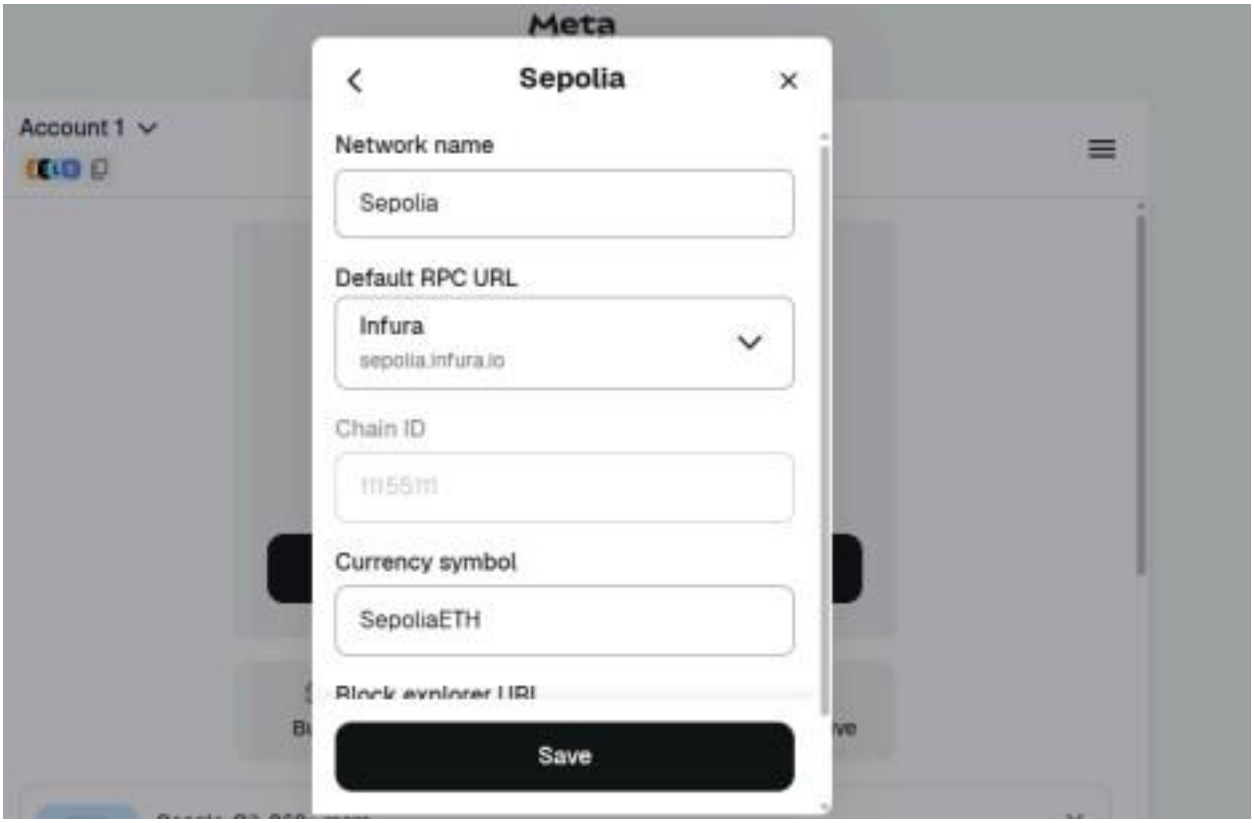
Step 3: Select Network (Testnet)

Choose a test blockchain:

Use Sepolia Testnet

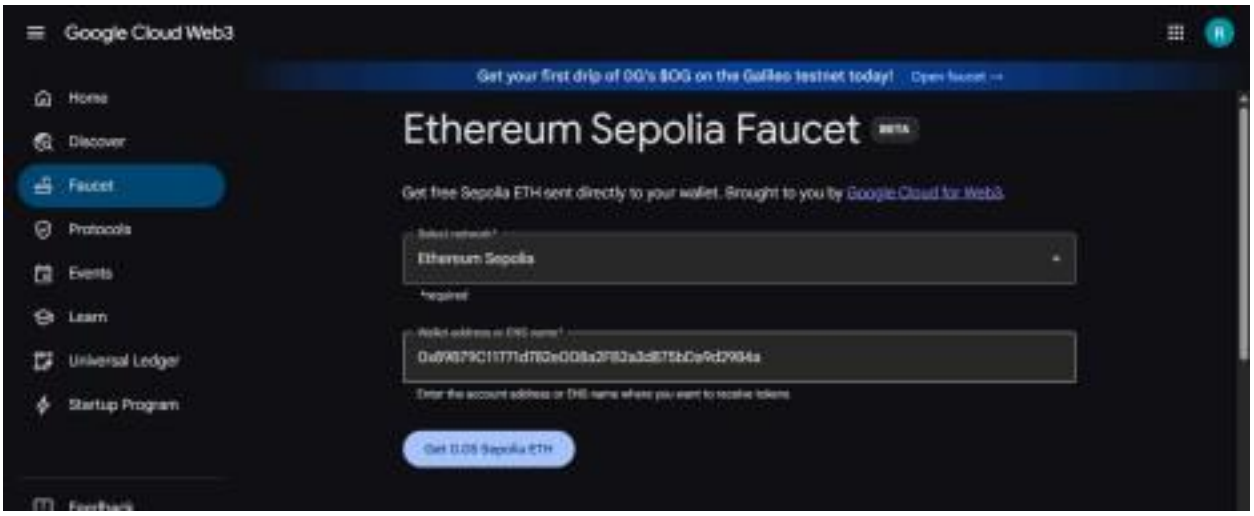
- In MetaMask → Network dropdown
- Select **Sepolia Testnet**

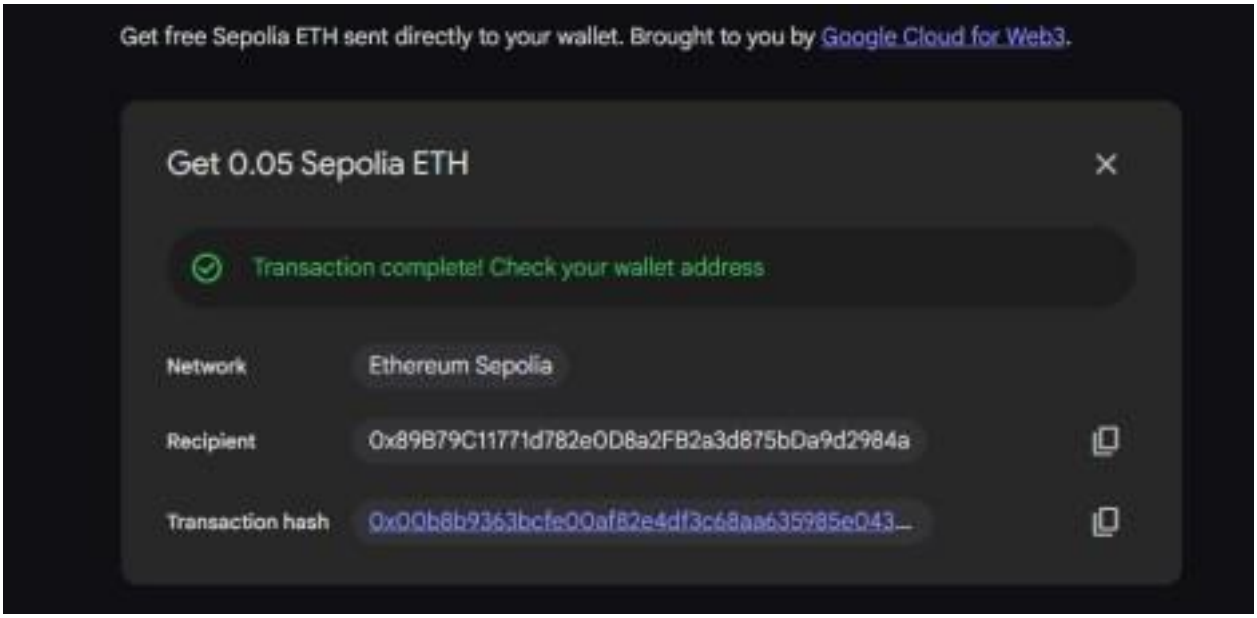


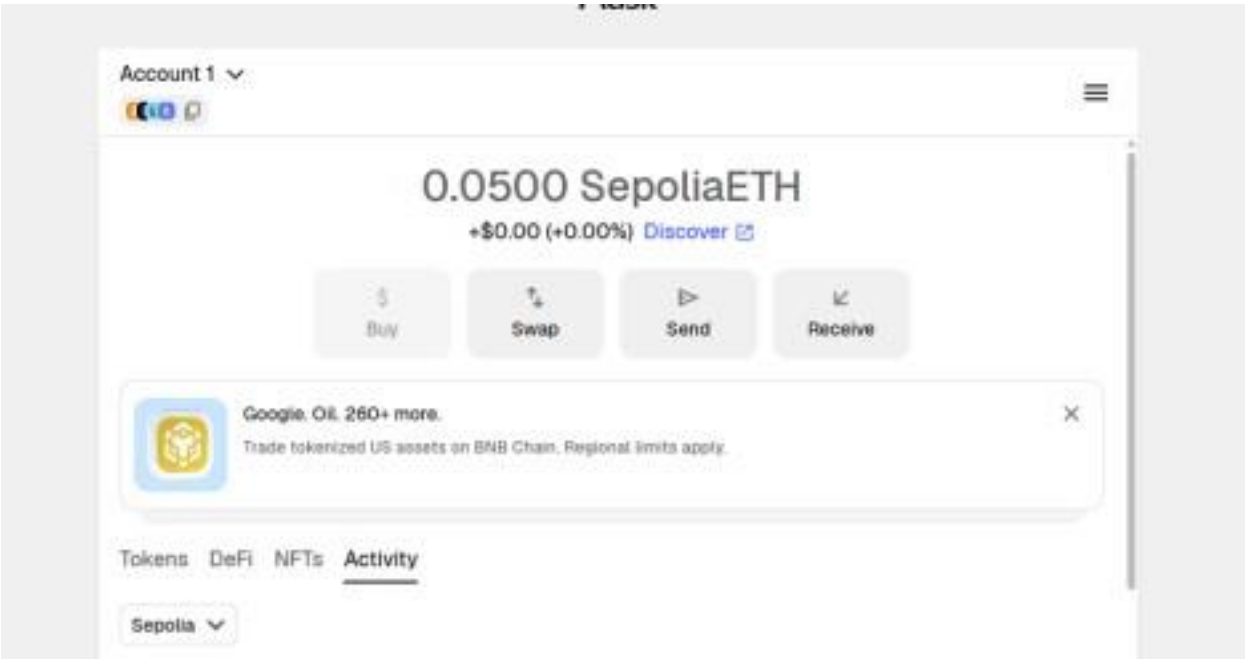


Step 4: Fund Your Wallet

- Copy your MetaMask address
- Go to Sepolia faucet
- Request test ETH
- Free ETH is added to your wallet

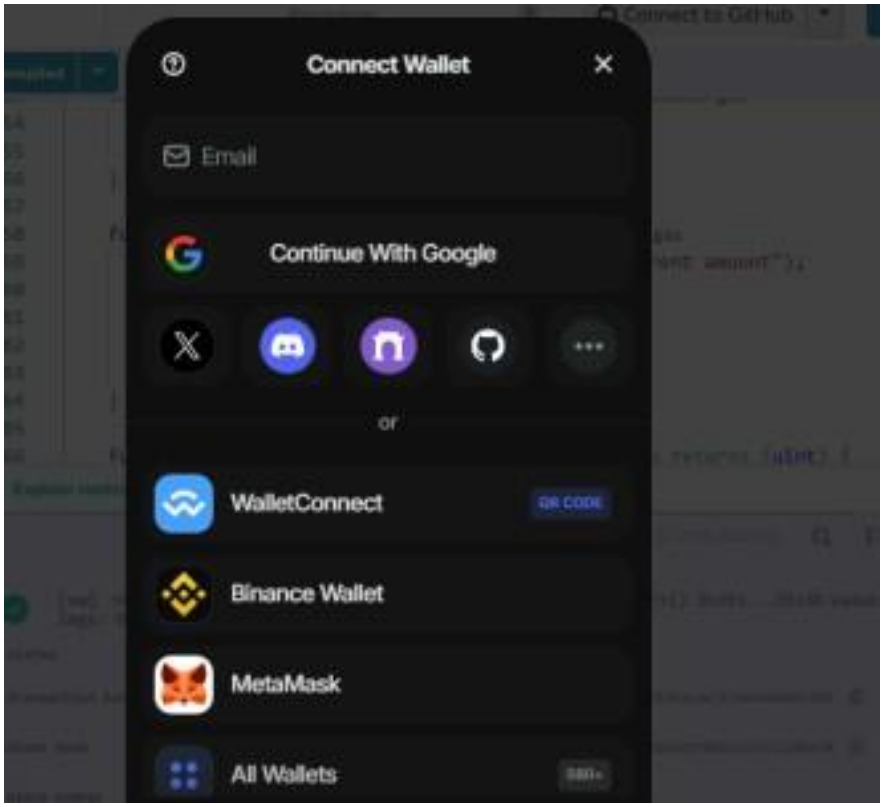


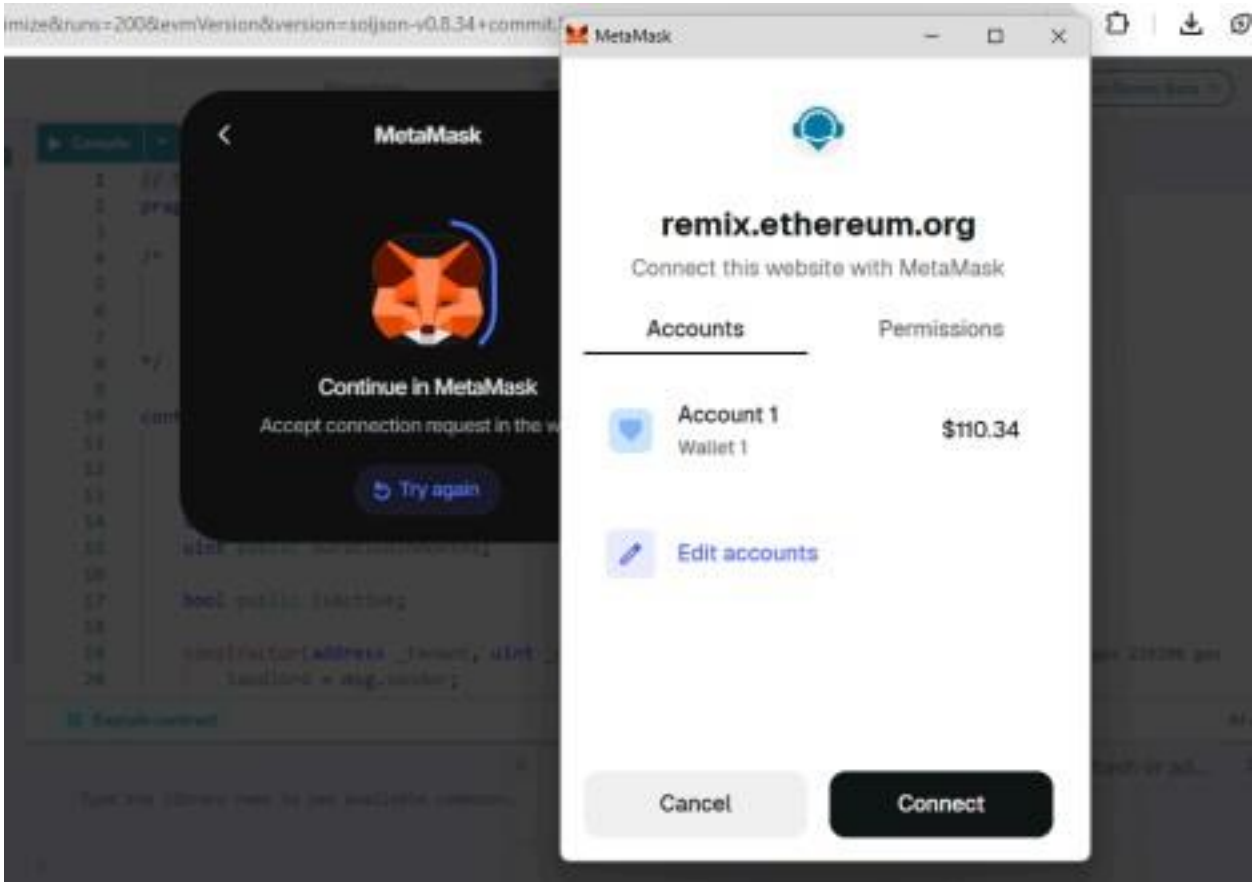




Step 5: Open and Connect Remix IDE

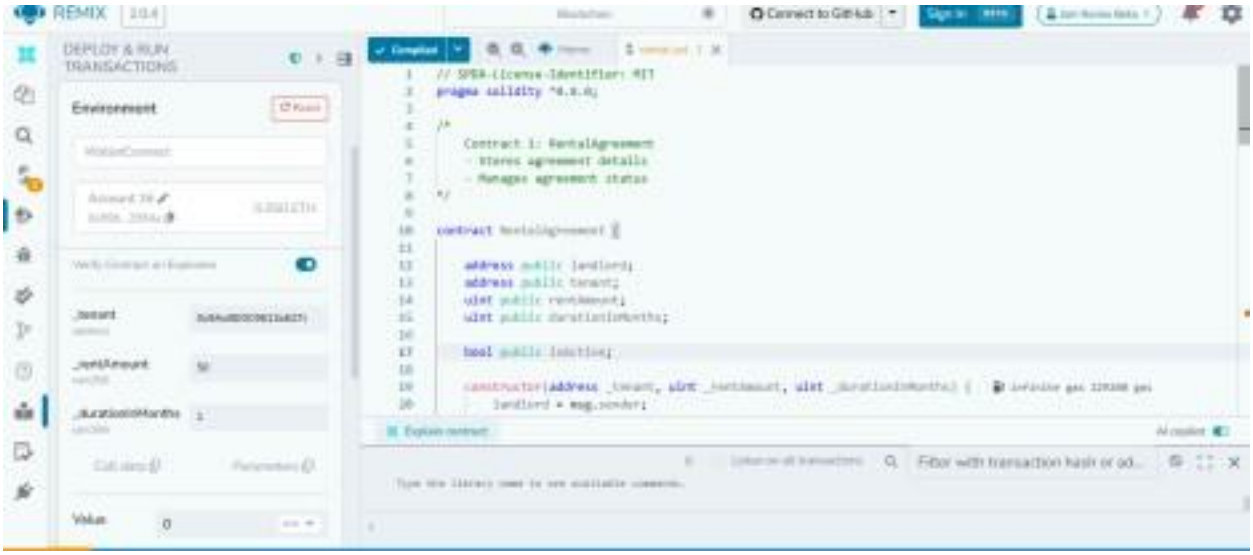
1. Open Remix
2. Go to **Deploy & Run Transactions**
3. Select: Injected Provider - MetaMask
4. Click **Connect** in MetaMask popup

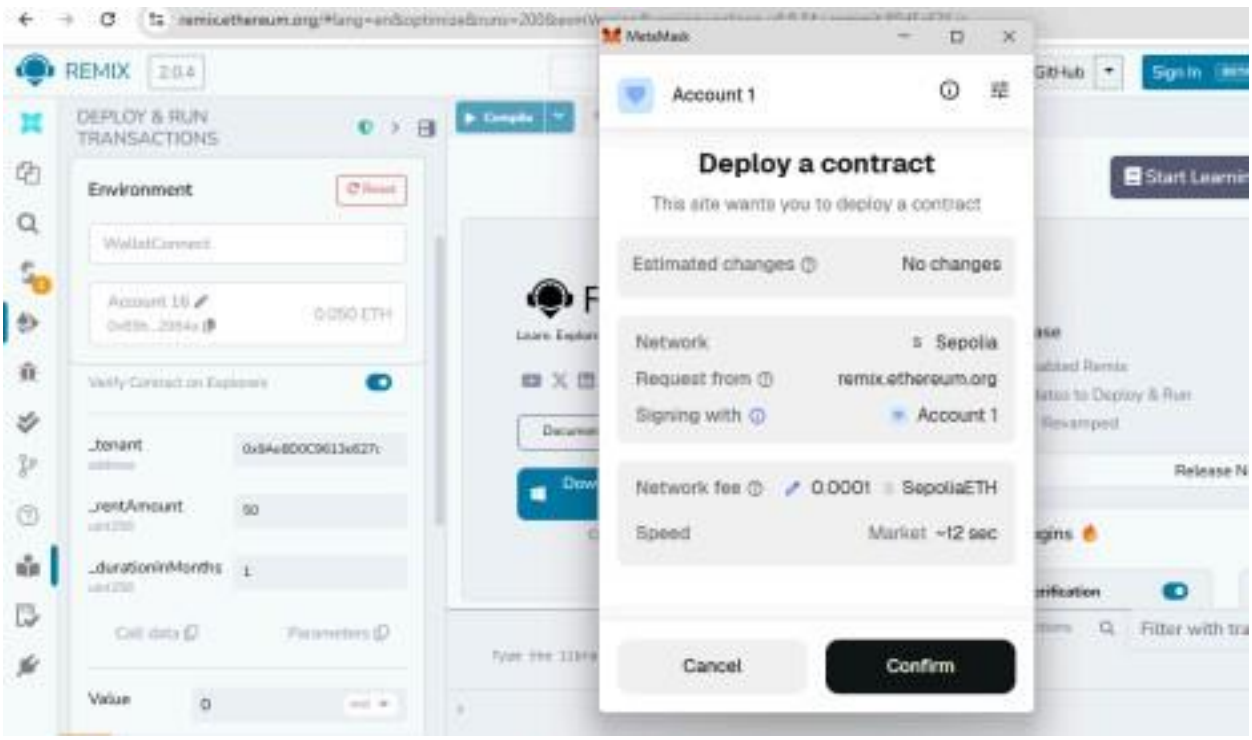


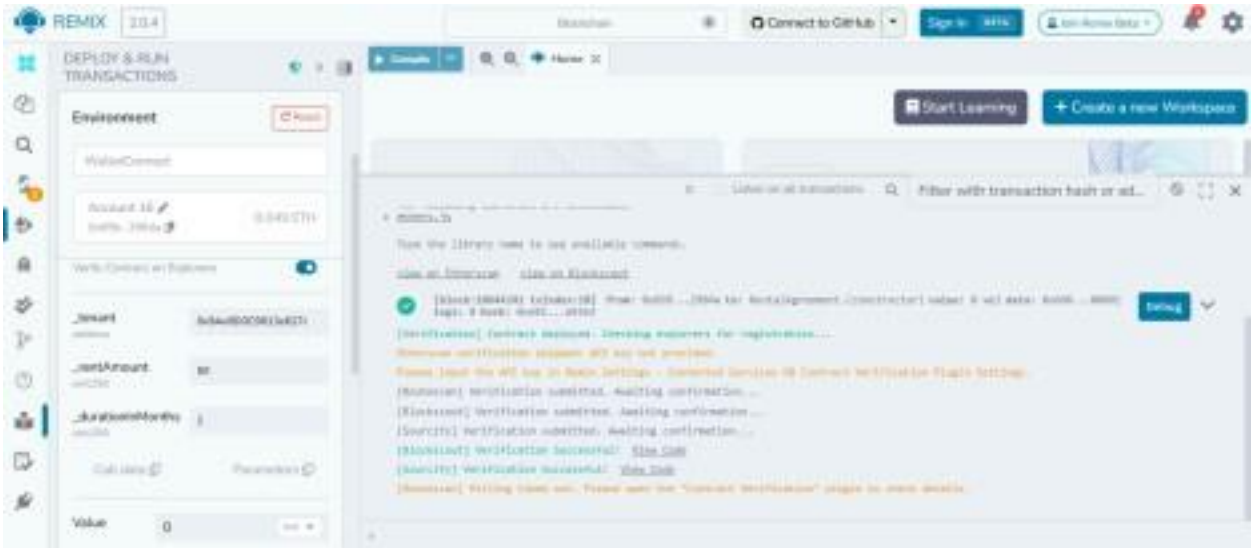


Step 8: Deploy RentalAgreement Contract

Tenant Address - Account 2
Rent Amount -100(in wei)
Duration(months)-1

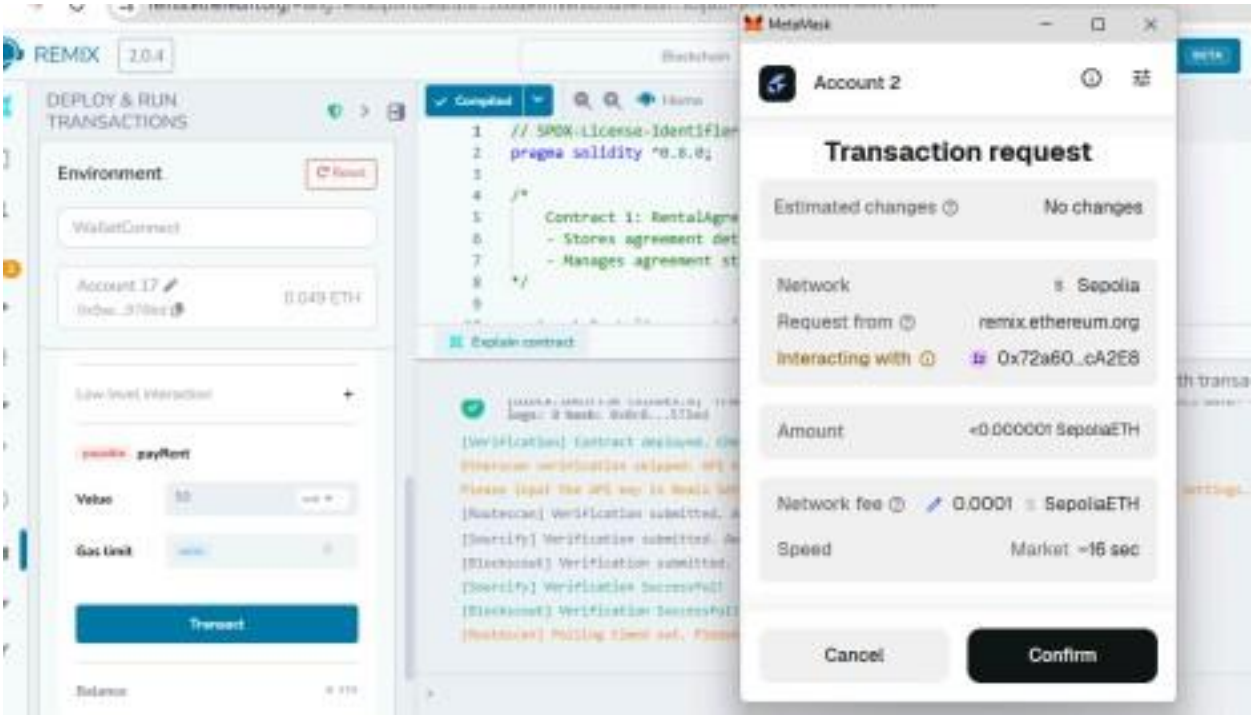




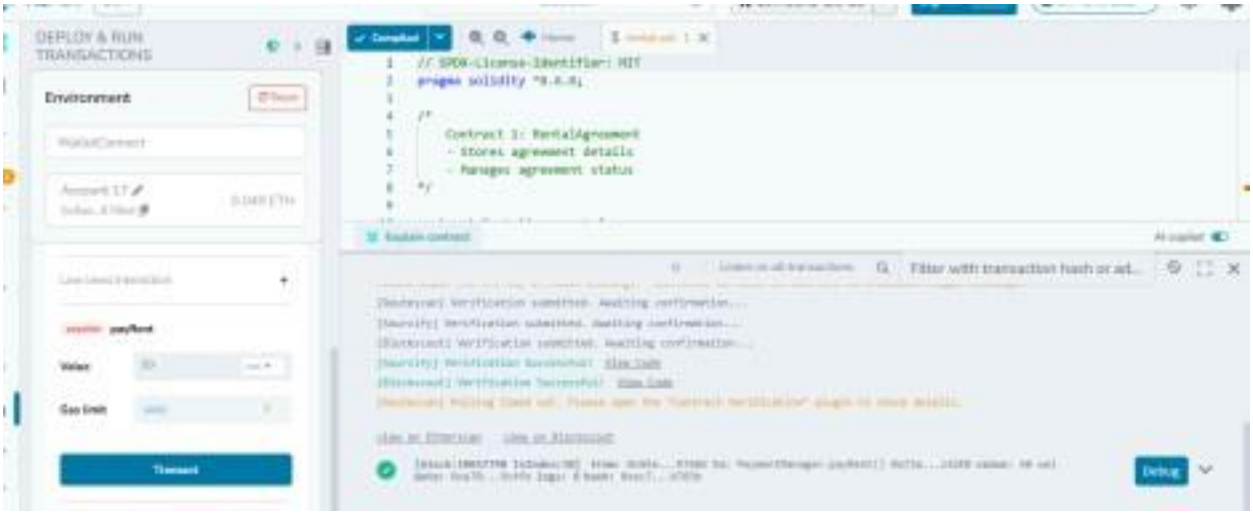


Step 9: Deploy PaymentManager Contract

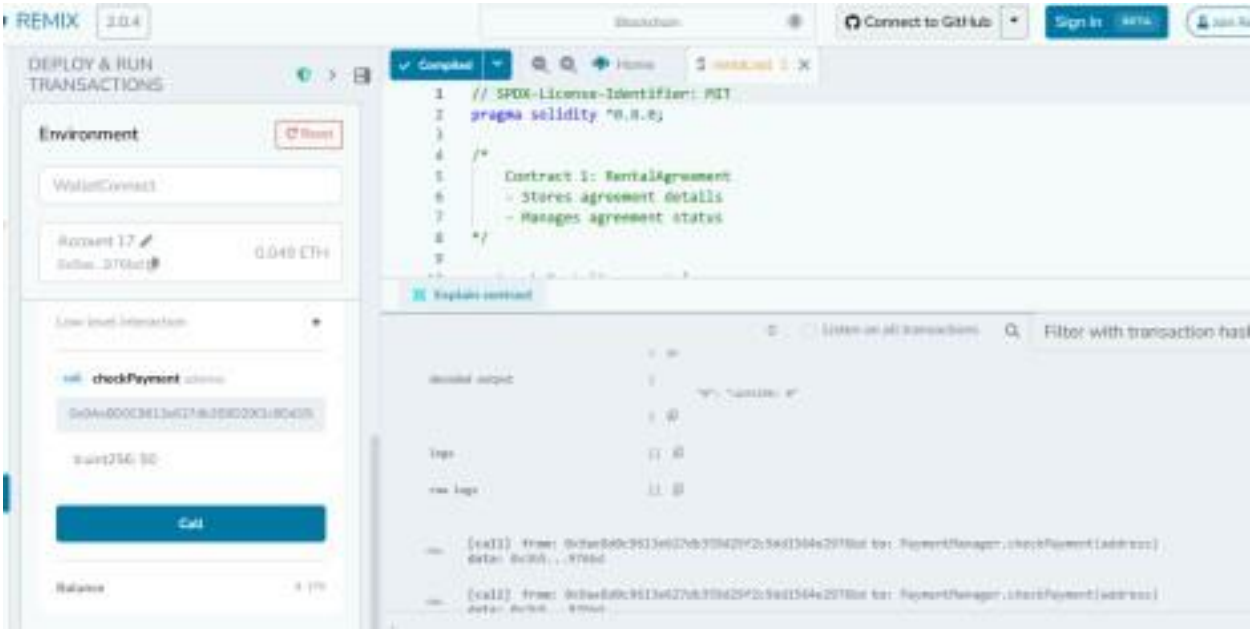
- Select Account 2
- Click: payRent()
- Rent Amount = 50
- Confirm in MetaMask



Amount transferred to landlord



Check payment



Step 11: Check Transaction on Blockchain Explorer

- Copy transaction hash from MetaMask
- Open: transaction list

Conclusion

The implementation of the Blockchain-based Rental Agreement System was successfully carried out by integrating MetaMask with Remix IDE and deploying smart contracts on the Sepolia Testnet. This setup enabled real-time interaction with the blockchain, allowing secure and authenticated execution of transactions.

Through MetaMask, users were able to create accounts representing the landlord and tenant, manage testnet funds, and sign transactions securely. The smart contracts were compiled and deployed using Remix, and rent payment transactions were successfully executed from the tenant account to the landlord account. Each transaction was validated on the blockchain and verified using a blockchain explorer, ensuring transparency and immutability.